Food Remedy API

Competitive App Research: Foodvisor [RE013]

Prepared For: Food Remedy API Team

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Introduction

Foodvisor is a mobile application centred on health and nutrition, that uses artificial intelligence to simplify meal tracking and provide nutritional guidance. Its standout feature is photo-based logging, which enables users to snap a picture, scan a barcode, search for food items, or use voice input to quickly receive a detailed nutritional breakdown. As the Food Remedy API team focuses into visual food recognition and allergen alert detections, Foodvisor serves as a valuable case study in AI integration, multi-modal food tracking, and the challenge of offering user-friendly features while restricting advanced tools behind a paywall.

This research explores Foodvisor's user interface, the accuracy of its food recognition capabilities, search functionality, and the impact of its premium features limitations. It also evaluates the app's relevance in shaping the development of an image-based allergen detection system.

Overview of the Foodvisor App

Foodvisor app is available for both iOS and Android platforms and is known for smart-nutrition guide. The app is designed for users who want to track their daily meals, monitor health goals, and receive the AI-driven dietary insights.

Key Features includes:

- Photo-based food logging using AI
- Barcode scanning
- Voice-based and manual food search
- Custom recipe creation
- Daily activity and weight tracking
- Access to healthy recipes (Premium Access).

User Interface and Registration Experience

Foodvisor UI is clean, intuitive, and visually appealing, offering a welcoming experience for new users. It uses a breadcrumb-style registration process, which guides users step-by-step through:

- Name and basic info
- Weight and environment status
- Health goals
- Meal preferences

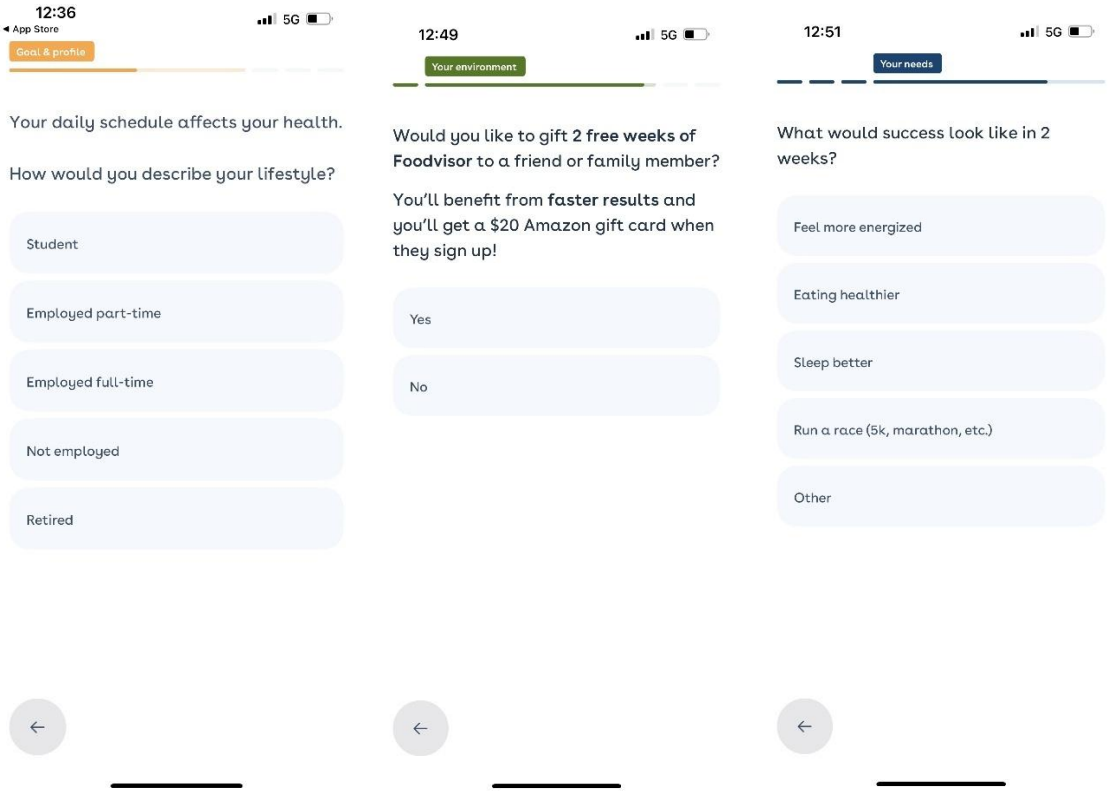

Figure 1: Different Breadcrumb registration process

While initially user-friendly, this process includes many consecutive steps, often asking overlapping or repetitive questions. This can cause friction for users seeking a quick setup, particularly when multiple breadcrumb pages must be completed before using core features.

Photo-Logging and Food Recognition

Photo-Logging Workflow:

- Users can log meals by taking a photo, scan barcode, search food, or uploading one from their gallery.
- The app identifies food items and provides estimated portions and nutrition information.

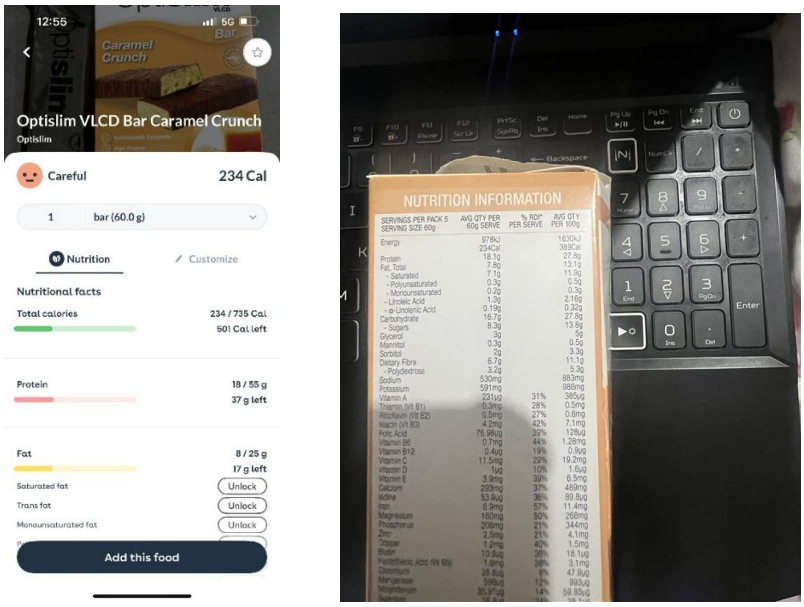

Figure 2: Accurate Nutritional Information

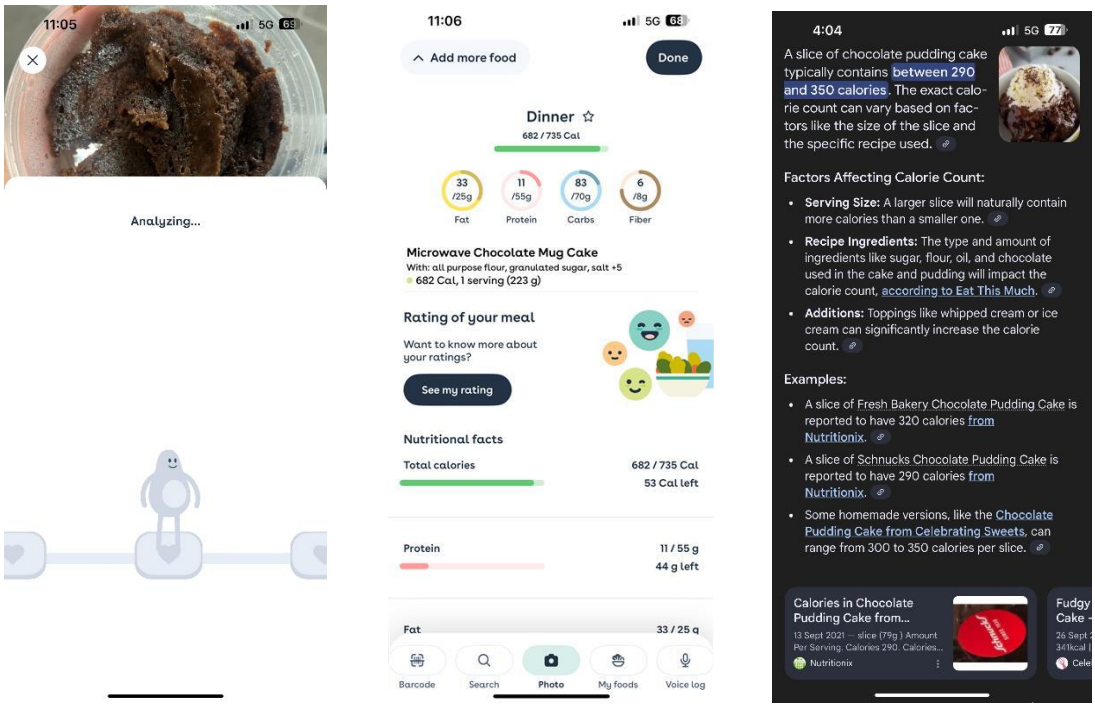

Figure 3: Searching calories using camera feature

Performance:

- **Accuracy**: The AI was reasonably accurate with common foods as shown in Figure 2.
- **Limitations**: The system tends to struggle with complex dishes and multi-item meals, such as mixed salads, curries, or cakes, where ingredient identification becomes more difficult as shown in figure 3. This may lead to misidentification or inaccurate portion estimation, which poses challenges for users with dietary restrictions or allergies.
- **Feedback**: Nutritional feedback is presented with emojis like careful and color-coded flags, providing quick visual interpretation (e.g., red for high fat/sugar).

These limitations highlight the importance of improving ingredient-level detection for allergen scanning use cases.

Food Search Capabilities

Foodvisor allows food entry through multiple input methods: (Figure 3)

- Barcode scanning for packaged food products.
- Manual search by entering food names.
- Using food photo from gallery.
- Voice-based search, which returns nutritional information (e.g., “yoghurt”).
- Create a new recipe feature to manually build custom food items.

These varied search methods make Foodvisor highly accessible for different user preferences and situations.

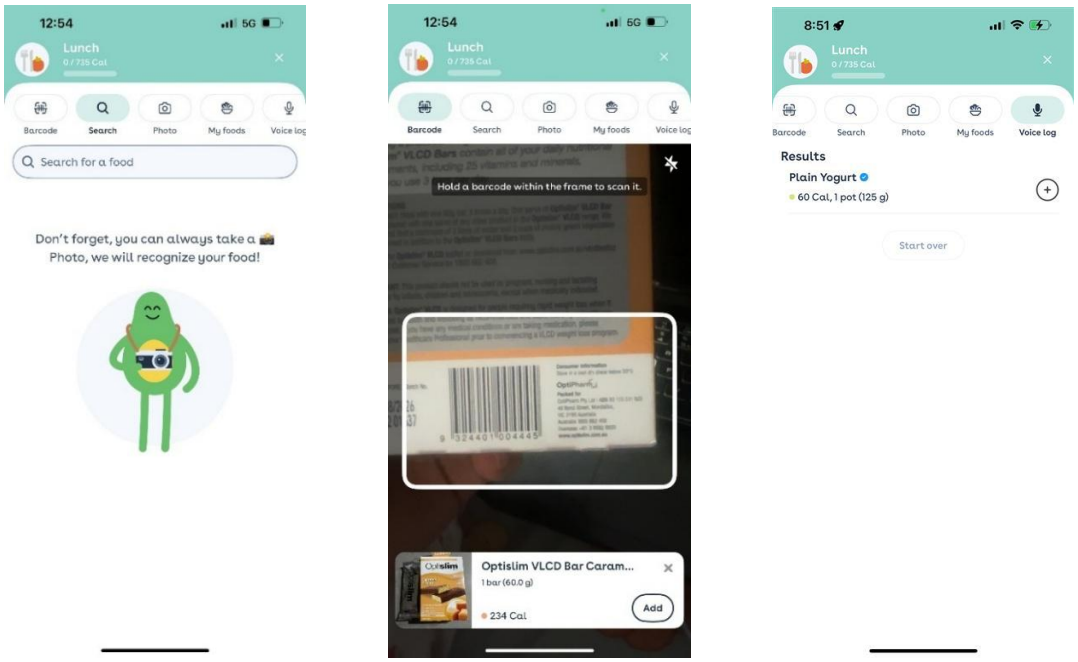

Figure 3: Different search capabilities of Foodvisor app

Premium Model and Feature Restrictions

The main limitation of Foodvisor is the premium paywall. Many advanced features are locked behind a subscription which is priced at approximately $68 AUD per year. This includes:

- Detailed micronutrient analysis (e.g., salt, calcium, vitamins, minerals) (Figure 4)
- Personalized coaching
- Advanced progress tracking

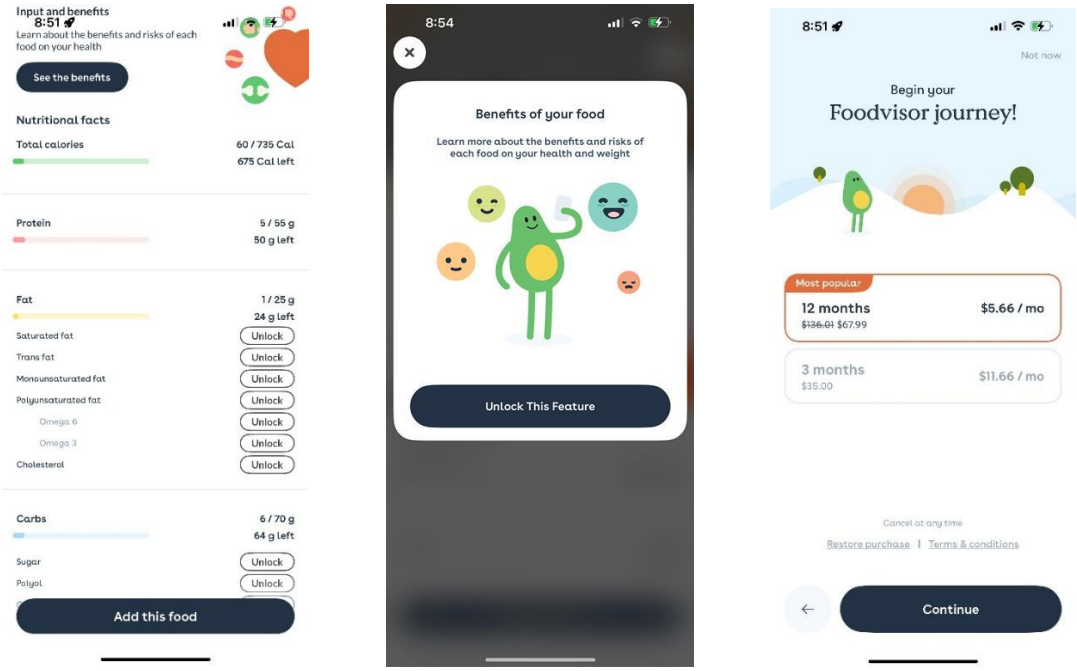

Figure 4: Premium features locked by the paywall

Free users can access only basic calorie and macronutrient data, limiting the app's usefulness for individuals with medical dietary needs such as allergies, vitamin deficiencies, or sodium restrictions.

Relevance to Food Remedy API – Allergen Detection

These Foodvisor features can be translated into allergy scan for Food Remedy API by:

- Using AI-Based Food Recognition to detect food types, and allergenic ingredients included in profile creation such as peanuts, dairy, gluten, or shellfish.
- Using feedback emojis and colours which can be adapted into warning flags or allergen icons for user clarity.
- Making personal health profiles, including allergens or intolerances fully accessible in the free version.
- Supporting barcode, photo, voice, and manual entry will broaden accessibility and improve data accuracy.

Conclusion

Foodvisor successfully integrates artificial intelligence, nutrition data, and user-friendly design to create a useful health tool. Its photo-logging tool provides precise nutritional information and offers a variety of search techniques. However, the app's AI struggles with mixed foods, which can result in inaccurate portion estimations or ingredient misidentification especially important for users with food allergies.

Its premium pricing approach limits access to critical health data, which the Food Remedy API addresses by providing fundamental allergen detection and nutritional insights for free. By analyzing Foodvisor's strengths and flaws, the Food Remedy API team can create a more inclusive, accurate, and health-focused platform with improved allergen alerts, multi-item recognition, and a more efficient registration process.